AI INDUSTRY IMPACT MAP

MINOR PROJECT 1 | ARTIFICIAL INTELLIGENCE

10 Real-world applications	**5** Industries mapped	**5+** AI method families

Research date: 19 October 2026

Evidence-led overview of AI applications, data requirements, business problems, impact metrics and responsible deployment.

Executive Summary

AI creates business value through pattern detection, prioritization, personalization, language generation and operational optimization. This report covers healthcare, finance, retail, education and transportation. It distinguishes peer-reviewed findings, company-reported metrics, platform usage statistics and benefits that still require local validation.

How to read the evidence

AUC, user counts, survey responses and operational KPIs measure different outcomes and should not be compared directly. Where a selected source does not provide a numerical business outcome, this report does not invent one.

Ten Real-World Applications

01 | Healthcare: AI-assisted chest X-ray analysis

AI methods: Deep Learning / Computer Vision
Business problem: Help clinicians detect and localize abnormalities in chest X-rays.
Data used: Chest X-ray pixels; annotated images; clinical labels.
Impact / evidence: A published evaluation reported AUC 0.976; physicians' AUC improved by 0.101 with AI assistance. Study result, not a guaranteed hospital-wide gain.

Source: https://pmc.ncbi.nlm.nih.gov/articles/PMC11502915/

02 | Healthcare: AI triage for urgent radiology scans

AI methods: Machine Learning / Computer Vision
Business problem: Prioritize potentially urgent scans for earlier review.
Data used: CT images; model scores; scan and report timestamps.
Impact / evidence: A systematic review found 67% of 33 studies measuring task time reported reductions, but pooled meta-analyses did not show significant overall effects.

Source: https://pubmed.ncbi.nlm.nih.gov/39349815/

03 | Finance: Real-time payment fraud detection

AI methods: Machine Learning; GenAI-enhanced scoring
Business problem: Flag suspicious transactions while limiting unnecessary declines.
Data used: Transaction amount/time; merchant; account/device signals; fraud outcomes.
Impact / evidence: Mastercard reported a 20% average improvement in initial fraud-detection modelling; company-reported estimate, not a universal independent benchmark.

Source: https://www.mastercard.com/in/en/news-and-trends/press/2024/february/mastercard-supercharges-consumer-protection-with-gen-ai.html

04 | Finance: Fraud investigation and case triage

AI methods: Machine Learning / NLP
Business problem: Rank suspicious cases and summarize signals to reduce manual investigation effort.
Data used: Transaction histories; alerts; investigator notes; customer and merchant context.
Impact / evidence: Mastercard's research reports 83% of surveyed organizations said AI significantly sped up investigation and resolution; survey result, not a causal experiment.

Source:
https://www.mastercard.com/global/en/news-and-trends/Insights/2026/ai-is-helping-banks-save-millions-by-transforming-payment-fraud-prevention.html

05 | Retail: Personalized product recommendations

AI methods: Machine Learning / Recommender Systems
Business problem: Help customers discover relevant products and improve shopping relevance.
Data used: Clicks; searches; purchases; product metadata; customer preferences.
Impact / evidence: Amazon describes AI-powered personalization and recommendation improvements; the cited source does not specify a public numerical revenue lift.

Source: https://www.aboutamazon.com/news/retail/amazon-generative-ai-product-search-results-and-descriptions

06 | Retail: AI-generated personalized product descriptions

AI methods: Generative AI / NLP
Business problem: Adapt descriptions to shopping context and highlight relevant product attributes.
Data used: Product catalog text; product attributes; search and shopping activity.
Impact / evidence: Potentially improves relevance and discovery; the cited source describes the capability but does not quantify a measured uplift.

Source: https://www.aboutamazon.com/news/retail/amazon-generative-ai-product-search-results-and-descriptions

07 | Education: AI tutor and teaching assistant (Khanmigo)

AI methods: Generative AI / NLP
Business problem: Provide guided explanations and help teachers draft learning materials and feedback.
Data used: Student prompts; exercise context; curriculum content; learning objectives.
Impact / evidence: Khan Academy reported 2.0 million Khanmigo users during school year 2024–25. Reach is not proof of causal learning gains.

Source: https://annualreport.khanacademy.org/

08 | Education: Adaptive practice and skill recommendations

AI methods: Machine Learning / Predictive Analytics
Business problem: Recommend practice based on learner progress and skill mastery.
Data used: Answers; attempts; time on task; skill mastery; assessment results.
Impact / evidence: Khan Academy reports broader platform research associating 20% learning gains with 18+ hours of practice in a school year; not attributable to GenAI alone.

Source: https://annualreport.khanacademy.org/

09 | Transportation: AI-assisted autonomous driving

AI methods: Deep Learning / Computer Vision / Sensor Fusion
Business problem: Perceive road users, lanes and traffic signals and predict motion.
Data used: Camera images; lidar/radar where used; maps; vehicle motion; labeled driving scenes.
Impact / evidence: Potential value includes driver assistance and consistent perception. Performance should be assessed with transparent safety metrics; no universal savings figure is claimed.

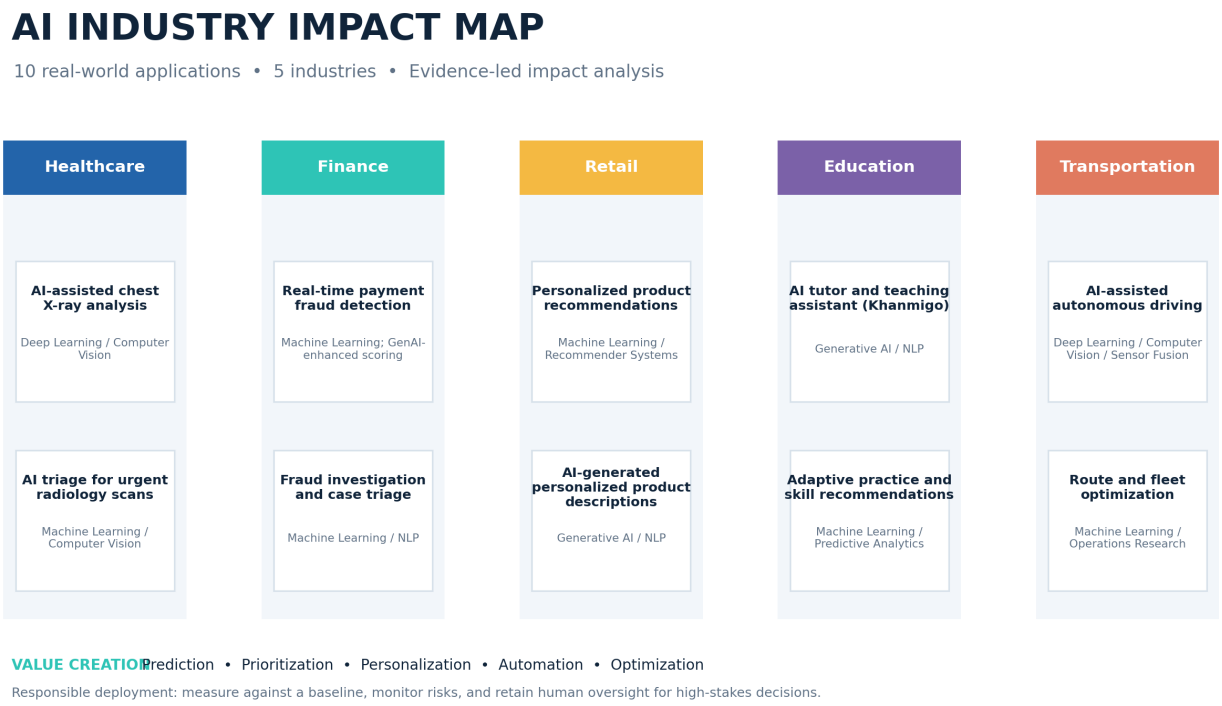
Source: https://www.who.int/teams/digital-health-and-innovation/harnessing-artificial-intelligence-for-health

10 | Transportation: Route and fleet optimization

AI methods: Machine Learning / Operations Research
Business problem: Plan efficient routes and dispatch schedules under time, capacity and traffic constraints.
Data used: Delivery locations; road network; travel times; vehicle capacity; time windows; route outcomes.
Impact / evidence: Measure miles per delivery, fuel use, on-time rate and cost per stop. A verified numerical impact was not found in the selected sources.

Source: https://www.who.int/teams/digital-health-and-innovation/harnessing-artificial-intelligence-for-health

Industry-wise AI Mapping



Cross-Industry Findings

- **Prediction and prioritization:** useful when data volume exceeds manual review capacity.
- **Personalization:** uses behavioral and contextual data to tailor recommendations or practice.
- **Generative AI:** supports language-heavy tasks but requires grounded content, guardrails and review.
- **Measurement:** run a local pilot before claiming savings or productivity gains.
- **Responsible deployment:** address privacy, bias, drift, hallucination, security and overreliance.

Recommended Measurement Plan

- Set one primary KPI and 2–3 guardrail metrics before deployment.
- Record a pre-AI baseline; compare with a controlled or matched pilot where feasible.
- Segment outcomes to detect uneven performance.
- Track false positives/negatives, human overrides, latency, operating cost and incidents.
- Review data permissions, retention, security and model drift regularly.

Conclusion

AI can create value through prediction, personalization, automation and decision support. The business case is strongest when the workflow is well-defined, data is reliable, impact is measurable and human accountability remains explicit. Not every application has a published, attributable financial return.

References

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